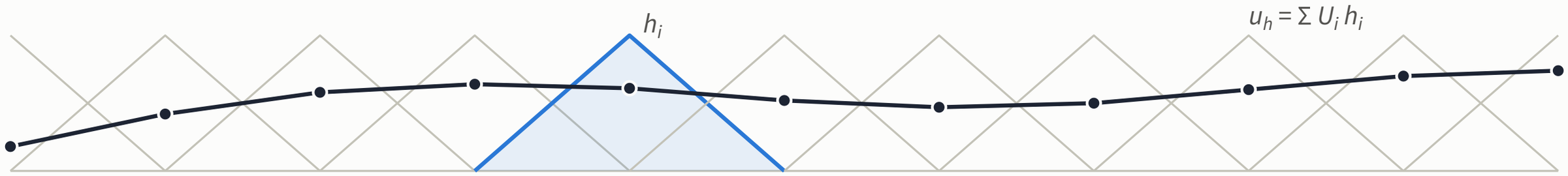
APPLIED MATH NOTES · CHAPTER 1

The Galerkin method

From the operator equation to a finite element solution in FEniCS



From a differential equation to a FEniCS solution

OPERATOR EQUATION $Au = f$

↓ multiply by a test function v, integrate over Ω

INTEGRAL FORM $(Au, v) = (f, v)$

↓ integrate by parts s times, insert the boundary conditions

VARIATIONAL FORMULATION $a(u, v) = L(v)$

↓ Galerkin method: $u_h = \sum_i U_i \, h_i$, test with $v = h_j$, solve $K\,U = F$

FENICS

```
solve(a == L, u, bcs)
```

Along the way, the order of A decides how often we integrate by parts, and which boundary conditions are ● **essential** and which are ▬ **natural**.

Inner product and operator equation

Inner product

of two functions on the domain Ω

$$(f, g) = \int_\Omega f\, g\, dV$$

Operator equation

a compact way to write a differential equation

$$A\, u = f, \qquad u \in D(A)$$

- A contains all operations applied to the unknown function $u = u(x)$, for example $A = -\frac{d^2}{dx^2} + 1$.
- The **order** $O(A)$ is the degree of the highest derivative in A. We write $O(A) = 2s$, where s is the number of integrations by parts that can be applied to the operator.

Two kinds of boundary conditions

The order of the operator decides the kind. With $O(A) = 2s$ we introduce $m = s - 1$.

Essential or Dirichlet

The order of the derivative prescribed on the boundary is at most m:

$$\frac{d^{(c)}u}{dx^{(c)}} \text{ prescribed on } \partial\Omega, \qquad c \leq m$$

In FEniCS: imposed on the function space with `DirichletBC`.

Natural or Neumann

The order of the derivative prescribed on the boundary is greater than m:

$$\frac{d^{(c)}u}{dx^{(c)}} \text{ prescribed on } \partial\Omega, \qquad c > m$$

In FEniCS: a boundary integral `... * ds` in the linear form.

Integral form of a differential equation

Multiply $Au = f$ by a **test function** v and integrate over the domain Ω:

$$\int_\Omega Au\, v\, dV = \int_\Omega f\, v\, dV \qquad \text{or} \qquad (Au, v) = (f, v)$$

- **Dirichlet boundary**
  where u is prescribed: $v = 0$

- **Neumann boundary**
  v is not subject to any condition

v is an arbitrary continuously differentiable function. Every solution of the operator equation also solves the integral form, for *every* choice of v.

Variational formulation

Integrate the integral form by parts s times and incorporate the boundary conditions:

$$\underbrace{(Au,\, v)}_{2s \text{ derivatives on } u} \xrightarrow[\text{insert boundary conditions}]{\text{integrate by parts } s \text{ times}} \underbrace{a(u,\, v)}_{s \text{ on } u,\ s \text{ on } v} = L(v)$$

- s derivatives move from u onto the test function v.
- Boundary terms appear. They vanish where $v = 0$ (Dirichlet boundary); on the Neumann boundary the prescribed data is inserted.

Symmetry, positivity and linearity of an operator

Symmetric

$$(Au, v) = (Av, u)$$

for all $u, v \in D(A)$

Positive

$$(Au, u) > 0 \quad \text{for } u \neq 0$$

and $(Au, u) = 0$ for $u = 0$, for $u \in D(A)$

Linear

$$A(u - v) = Au - Av$$

The Galerkin method

Exact solution

The solution u satisfies the variational formulation for *arbitrary* test functions:

$$a(u, v) = L(v) \qquad \text{for all } v$$

infinitely many conditions

Idea: a finite approximation

Approximate u with N basis functions h_i, and test with the same functions:

$$u_h = \sum_{i=1}^{N} U_i \, h_i, \qquad v = h_j$$

$V_h = \mathbf{span}\{h_1, \ldots, h_N\}$, finite-dimensional

Require the variational formulation for all test functions of V_h (testing with the h_j is enough, since a and L are linear in v):

$$a(u_h, h_j) = \sum_{i=1}^{N} U_i \, a(h_i, h_j) = L(h_j), \qquad j = 1, \ldots, N$$

N equations for the N coefficients U_i: the linear system $K\,U = F$ with $K_{ji} = a(h_i, h_j)$ and $F_j = L(h_j)$.

This is the Galerkin method. In general $u_h \neq u$, but u_h is at least a weak solution with respect to V_h: the variational formulation holds for every $v_h \in V_h$.

EXAMPLE

Heat conduction: $-\nabla \cdot \nabla u = f$ in Ω

Operator

$$A = -\nabla \cdot \nabla$$

$$O(A) = 2, \quad s = 1, \quad m = 0$$

- **Dirichlet** $c = 0 \leq m$
  u is prescribed on the boundary

- **Neumann** $c = 1 > m$
  the normal derivative $\nabla u \cdot n$ is prescribed (n: outward normal vector)

Integral form

$$(Au, v) = -\int_\Omega (\nabla \cdot \nabla u)\, v \, dx = \int_\Omega f\, v \, dx = (f, v)$$

EXAMPLE

Integration by parts, once since $s = 1$

Gauss's divergence theorem:

$$-\int_\Omega (\nabla \cdot \nabla u)\, v \, dx = \int_\Omega \nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} (\nabla u \cdot n)\, v \, ds$$

The boundary term vanishes where ● $v = 0$ (Dirichlet boundary) or where ▬ $\nabla u \cdot n = 0$ (homogeneous Neumann boundary).

Symmetric

$$(Au, v) = \int_\Omega \nabla u \cdot \nabla v \, dx = (Av, u)$$

Positive

$$(Au, u) = \int_\Omega |\nabla u|^2 \, dx \geq 0$$

for $u, v \in D(A)$ with homogeneous boundary conditions, so that the boundary term vanishes

EXAMPLE

Variational formulation with a Neumann boundary Γ

On $\Gamma \subset \partial\Omega$ the outward heat flux is prescribed, $q \cdot n = g$ with $q = -\nabla u$ (Fourier's law, conductivity 1). On $\partial\Omega \setminus \Gamma$, u is prescribed and $v = 0$. Find u such that

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f \, v \, dx - \int_\Gamma g \, v \, ds \qquad \text{for all } v.$$

$a(u, v)$

$$\int_\Omega \nabla u \cdot \nabla v \, dx$$

```
inner(grad(u), grad(v)) * dx
```

$L(v)$ · source

$$\int_\Omega f \, v \, dx$$

```
f * v * dx
```

$L(v)$ · Neumann boundary

$$-\int_\Gamma g \, v \, ds$$

```
- g * v * ds
```

Exactly this form is used in the FEniCS examples: the left-hand side in examples 1 to 3, the boundary term in example 3.

Shared setup: mesh, function space, boundary markers

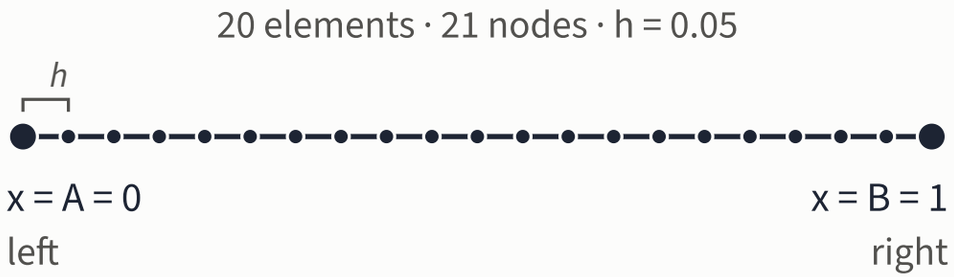
```
from fenics import *

tol = 1e-3
A   = 0.0
B   = 1.0
nr_elements = 20

mesh = IntervalMesh(nr_elements, A, B)
s    = FiniteElement('Lagrange', mesh.ufl_cell(), 1)
V    = FunctionSpace(mesh, s)

def left(x, on_boundary):
    return on_boundary and near(x[0], A, tol)

def right(x, on_boundary):
    return on_boundary and near(x[0], B, tol)
```



All three examples solve a 1D heat conduction problem on $\Omega = (0, 1)$ with piecewise linear Lagrange elements, $-u'' = f$, and differ only in f and the boundary conditions.

Dirichlet – Dirichlet

Temperature fixed at both ends, no source term:

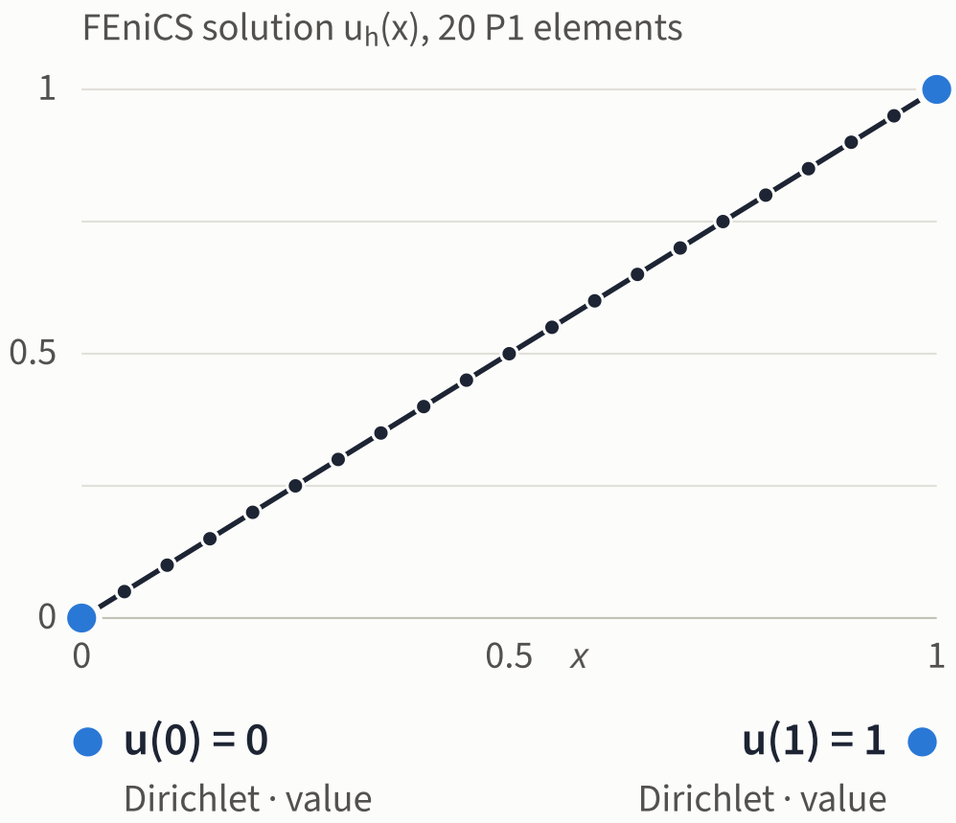
$$-u'' = 0, \qquad u(0) = 0, \quad u(1) = 1$$

```
bc0 = DirichletBC(V, Constant(0.0), left)
bc1 = DirichletBC(V, Constant(1.0), right)

u = TrialFunction(V)
v = TestFunction(V)
f = Constant(0.0)

a = inner(grad(u), grad(v)) * dx
L = f * v * dx

u = Function(V)
solve(a == L, u, [bc0, bc1])
plot(u)
```



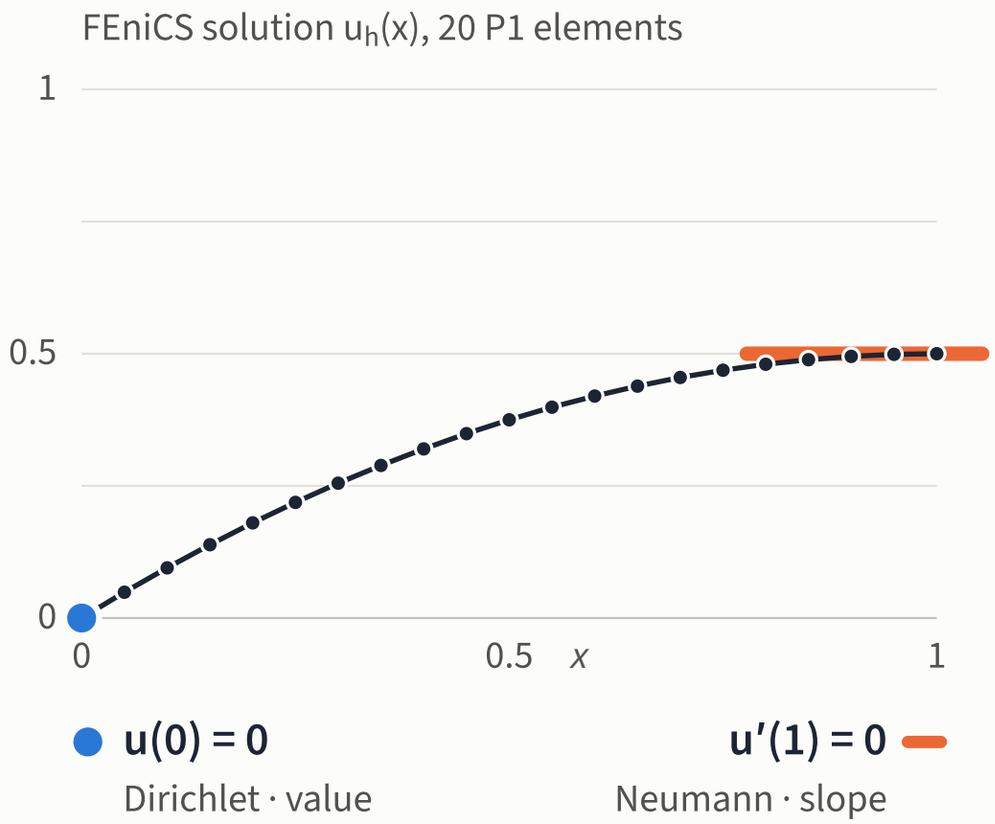
Dirichlet – homogeneous Neumann

Temperature fixed on the left, constant heat source, insulated right end:

$$-u'' = 1, \qquad u(0) = 0, \quad u'(1) = 0$$

```
1  bc = DirichletBC(V, Constant(0.0), left)
2
3  u = TrialFunction(V)
4  v = TestFunction(V)
5  f = Constant(1.0)
6
7  a = inner(grad(u), grad(v)) * dx
8  L = f * v * dx
9
10 u = Function(V)
11 solve(a == L, u, [bc])
12 plot(u)
```

Highlighted: the changes with respect to example 1. The right end gets no condition in the code at all.



Dirichlet – Neumann

Temperature fixed on the left, no source, flux $g = -1$ prescribed on the right:

$$-u'' = 0, \qquad u(0) = 0, \quad -u'(1) = g = -1$$

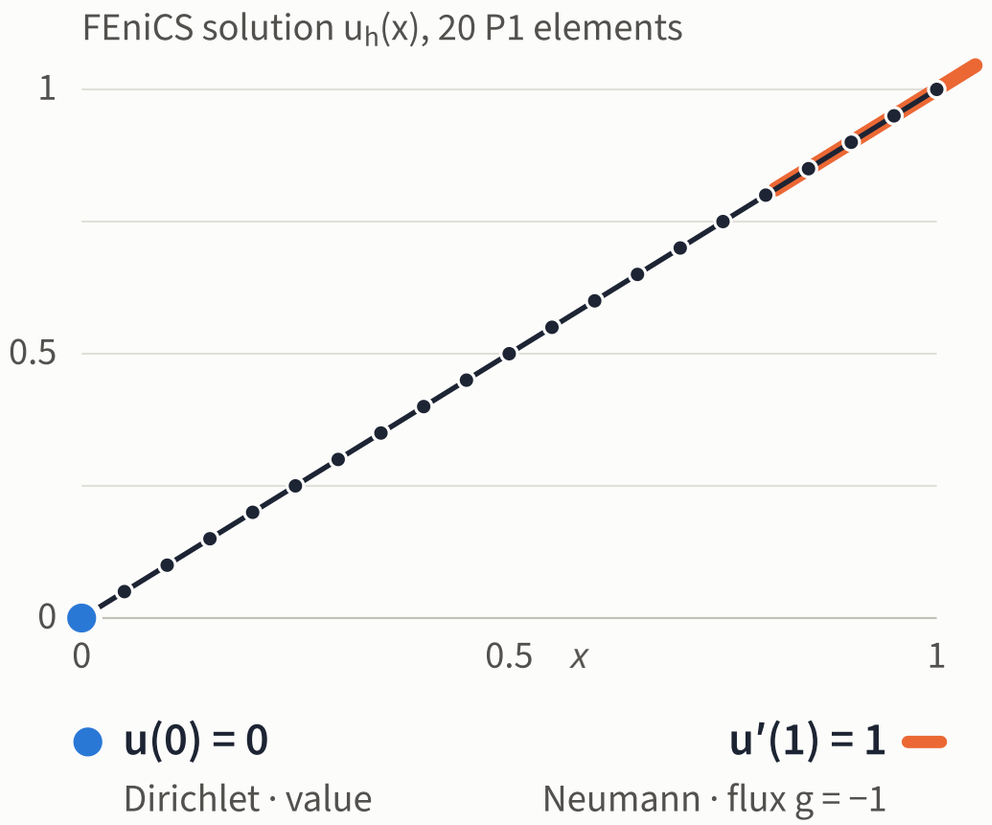
```
bc = DirichletBC(V, Constant(0.0), left)

u = TrialFunction(V)
v = TestFunction(V)
f = Constant(0.0)
g = Constant(-1.0)

a = inner(grad(u), grad(v)) * dx
L = f * v * dx - g * v * ds

u = Function(V)
solve(a == L, u, [bc])
plot(u)
```

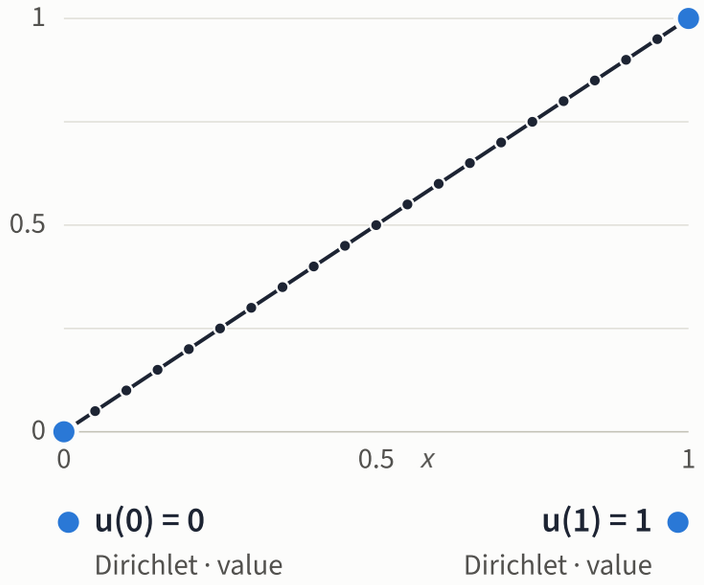
Highlighted: the Neumann data enters only through the boundary integral in `L`.



What the boundary conditions do

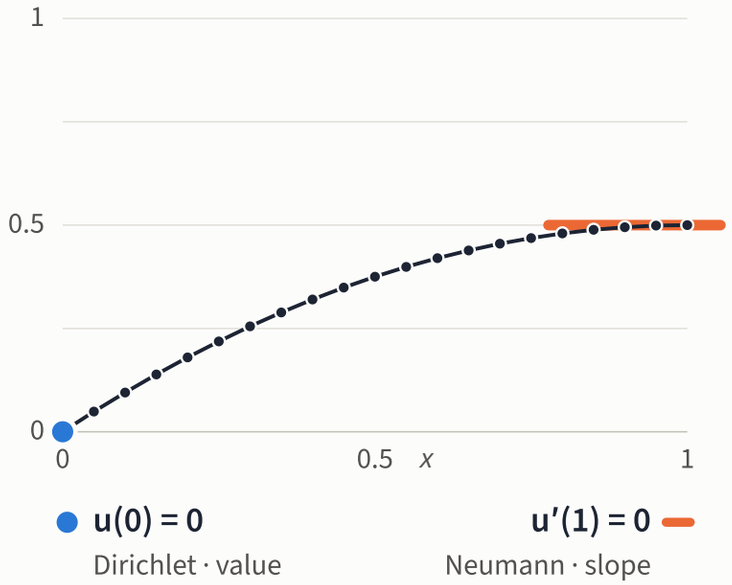
1 · Dirichlet – Dirichlet

$f = 0$



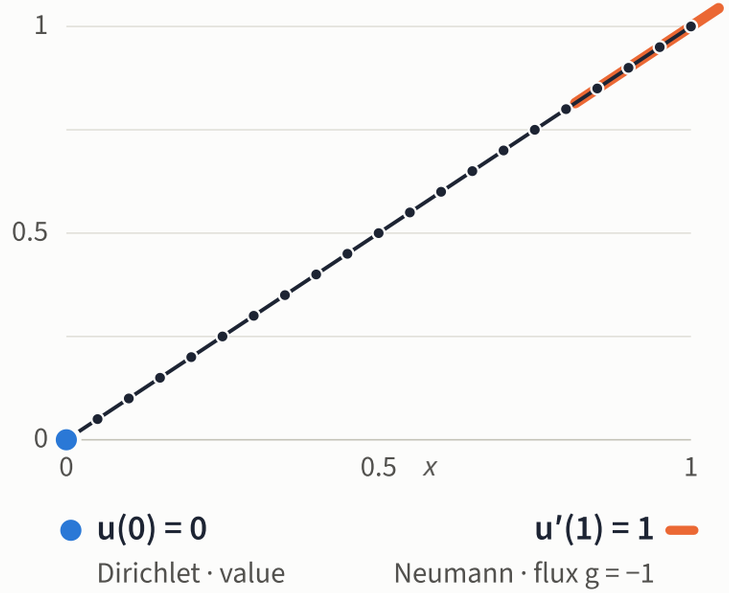
2 · Dirichlet – homogeneous Neumann

$f = 1$



3 · Dirichlet – Neumann

$f = 0,\ g = -1$



● **Dirichlet** prescribes the **value** of u · ▬ **Neumann** prescribes the **slope** u', i.e. the flux

The recipe

1. Write the problem as an operator equation $Au = f$ and determine $O(A) = 2s$ and $m = s - 1$.
2. Classify the boundary conditions: ● $c \leq m$: **Dirichlet**, built into the function space (`DirichletBC`); ▬ $c > m$: **Neumann**, built into the weak form (`... * ds`).
3. Multiply by a test function v, integrate over Ω, and integrate by parts s times: $a(u, v) = L(v)$.
4. Galerkin method: approximate $u_h = \sum_i U_i \, h_i$ and test with $v = h_j$. In FEniCS, write down `a` and `L` and call `solve(a == L, u, bcs)`, which assembles and solves $K\,U = F$.

Analytical solution and error norm for problem 2

Problem 2 reads: $-u'' = 1$ on $(0, 1)$ with $u(0) = 0$ and $u'(1) = 0$.

1. Derive the analytical solution $u(x)$ (integrate twice, insert the boundary conditions).
2. Solve the problem with FEniCS (Lagrange elements of degree 1) and compare the nodal values of the FEM solution with $u(x)$ at the nodes. What do you notice? Plot the error $u - u_h$ between the nodes.
3. Define an error norm, e.g. the L^2 norm $\|u - u_h\|_{L^2} = \left(\int_0^1 (u - u_h)^2 \, dx \right)^{1/2}$, and evaluate it with FEniCS (`assemble` or `errornorm`).
4. Increase the polynomial degree of the FEM approximation to 2 and 3 (`FunctionSpace(mesh, 'Lagrange', 2)`) and evaluate the error norm again. Explain the result.

Biharmonic equation (theory)

Given is the biharmonic equation in 1D on $\Omega = (0, 1)$ (beam bending: u is the deflection, f the distributed load):

$$Au = \frac{d^4 u}{dx^4} = f.$$

1. Determine the order $O(A)$ as well as s and m.
2. Set up the integral form $(Au, v) = (f, v)$ and derive the variational formulation by integrating by parts s times. Which boundary terms appear?
3. According to the criterion $c \leq m$ or $c > m$, which boundary conditions are Dirichlet (essential) and which are Neumann (natural) boundary conditions? Which conditions must the test function v satisfy?

Galerkin method with cubic Hermite trial functions

We set $f = 0$ and drop the boundary terms. The loading is instead prescribed directly as a load vector. Choose finitely many trial functions $h_i(x)$ and seek $u_h(x) = \sum_i U_i\, h_i(x)$ with unknown coefficients U_i. The same functions are used as test functions, $v_h = h_j$. Inserting into the variational formulation from item 2 gives, for each j,

$$\sum_i U_i \int_\Omega h_i'' \, h_j'' \, dx = \int_\Omega f \, h_j \, dx, \qquad \text{i.e.} \qquad K\,U = F$$

with $K_{ij} = \int_\Omega h_i'' \, h_j'' \, dx$ and $F_j = \int_\Omega f \, h_j \, dx$.

Trial functions: the cubic Hermite interpolation functions on $\Omega = (0, 1)$

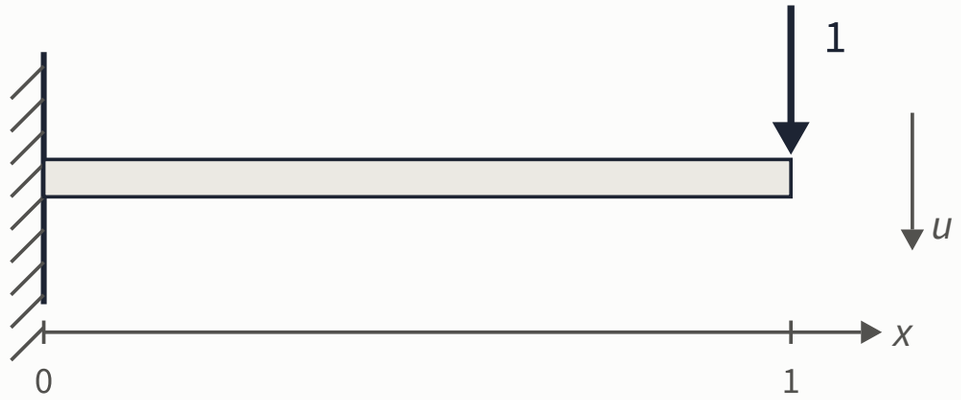
$$h_1 = 1 - 3x^2 + 2x^3, \quad h_2 = x - 2x^2 + x^3, \quad h_3 = 3x^2 - 2x^3, \quad h_4 = -x^2 + x^3$$

They are constructed such that $U_1 = u_h(0), U_2 = u_h'(0), U_3 = u_h(1), U_4 = u_h'(1)$. Check this using $h_1(0) = 1$ and $h_1'(0) = h_1(1) = h_1'(1) = 0$, and correspondingly for the others.

The cantilever beam

The beam is clamped at $x = 0$, $u(0) = 0$ and $u'(0) = 0$; the right end is free. The load vector for the free degrees of freedom $(U_3, U_4) = (u(1), u'(1))$ is

$$\underline{F} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$



i.e. a point force 1 at the free end and no moment.

a. Which of the h_j are admissible test functions? Which rows and columns of K are therefore needed?

b. Compute the required entries $K_{33}, K_{34}, K_{43}, K_{44}$. Why is $K_{34} = K_{43}$?

c. Solve the system of equations and give $u_h(x)$. Compare with the exact solution of $u'''' = 0$ with $u(0) = u'(0) = 0$, $u''(1) = 0$ and $u'''(1) = -1$.